

MANISH BIKUMALLA

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Software Developer with 3+ years of experience specializing in building high performance applications using Java, Spring Boot, and Python and providing ML and AI driven solutions. Full stack developer with experience in architecting scalable microservices and implementing responsive front-end interfaces with ReactJS. Committed to leveraging these skills to drive impactful business outcomes.

Technical Skills

Programming Languages: Python, Java, SQL, JavaScript (ES6+)

Front-End: ReactJS, HTML5, CSS, ChartJS, Redux

Back-End & Cloud Technologies: Spring, Spring Boot, Maven, MySQL, Microservices, REST APIs, Snowflake, AWS

Data & AI/ML: Numpy, Pandas, PyTorch, OpenCV, Langchain, LLM, RAG, Hadoop, Apache Spark, Streamlit, Flask

Tools & Methodologies: Docker, Microsoft Office Suite, Git, GitHub, Agile, Scrum, TDD

Coursework – Algorithms, Analysis and Design, Data Models and Query Languages, Deep Learning, Data Intensive Computing, Operating systems, Computer Vision, Predictive Analytics in Business

Education

University at Buffalo, State University of New York

January 2024 – June 2025

Master of Science in Computer Science and Engineering (GPA – 3.7/4)

IIT Kharagpur, India

B.Tech in Metallurgical and Materials Engineering (GPA – 3.6/4)

Experience

Software Developer, CBIC, India

November 2021 – January 2024

- Engineered a suite of **Java and Spring Boot** based **REST APIs** for tax evasion detection which resulted in up to 24% indirect tax revenue for the Financial Year 2022-23.
- Automated monthly brochure generation via a **dedicated microservice**, cutting production time by at least 40%.
- Engineered a real-time notification system using **Spring validation and Java**, alerting taxpayers about deficits in return filing errors, this led to reduced manual communication workload by 20% and improved compliance.
- Implemented **Elasticsearch with pagination** in the frontend using ReactJS for the Advait portal to enable fast, scalable retrieval of nearly 80,000 taxpayer records, empowering employees to perform advanced analytics and improve operational efficiency.
- Optimized application performance by **implementing lazy loading** techniques for rendering high-volume chart visualizations, reducing initial load time by 8%.
- Created 15+ **modular ReactJS components** for the Advait portal **and integrated Redux** for centralized state management to support a large-scale data rich user interface.
- Set up and maintained **CI/CD pipelines** with Jenkins and Docker reducing deployment time by 15%, enabling consistent and efficient software releases.

Data Analyst, Office of the AG, India

May 2020 – November 2020

- Leveraged SQL and Advanced Excel** to analyze state budgetary controls, identifying significant anomalies including a \$1.35 billion Revenue Surplus overstatement and \$500 million in unapproved spending.
- Automated **KPI compliance checks using Python**, validating Fiscal Deficit compliance 3.11% vs. 3.25% target and identifying that Interest Payments exceeded the national limit by 48%.
- Partnered with cross functional teams in an Agile environment across 10+ bi-weekly sprints, and continuously integrating stakeholder feedback to improve project outcomes.

Key Projects

Multimodal Emotion Detection using Computer Vision | *OpenCV, Transformers, OpenAI*

April 2025 – June 2025

- Streamlined facial feature extraction by detecting faces with OpenCV Haar Cascades and generating 512-D embeddings via OpenAI's CLIP Vision Transformer.
- Achieved a 15% uplift in emotion classification accuracy by fusing facial and pose embeddings within a chunked and cross-attention based custom Vision Transformer.

ChestAI | *Python, Pytorch, Deep Learning*

June 2024- July 2024

- Trained and fine-tuned a deep learning model using a custom CNN, ResNet and DenseNet models for chest X-ray classification with 92.85% accuracy on a custom dataset.
- Boosted model efficiency with GAN-based data augmentation and autoencoder-driven dimensionality reduction, reducing training time by 90%.

Achievements/Leadership

- Admitted to IIT Kharagpur based on a top 0.5% performance in India's highly competitive national engineering entrance exam.
- Prototyped an AI-driven solution in a 24-hour sprint at UB's AI Hackathon 2024, collaborating in a team of 3 to address a real-world challenge.
- Rated 9+ out of 10 in two consecutive annual performance appraisals for technical excellence and collaborative contributions.
- Spearheaded full-stack development for Community Dreams Foundation's new membership platform, leveraging Lovable and Supabase to enhance user experience and accommodate a 3,000-member increase.